3.4.3 - 3.4.4

Saturday, April 11, 2026 9:10 AM

Exercise 30. Consider the polynomial $P = x^7 + 3x^6 + 3x^5 + x^4 - x^3 - 3x^2 - 3x - 1 \in \mathbb{Z}_6[x]$. Compute the set of all roots of $R_0(P)$ and then compute the prime factorization of P.

Finding root of P in $\mathbb{Z}_6$

check each $a \in \{0, 1, 2, 3, 4, 5\}$

$P(0) = -1$ ×

$P(1) = 1^7 + 3(1)^6 + 3(1)^5 + (1)^4 - (1)^3 - 3(1)^2 - 3(1) - 1$

$= 1 + 3 + 3 + 1 - 1 - 3 - 3 - 1$

$= 0$ ✓

$P(2) = 2^7 + 3.2^6 + 3\,2^5 + 2^4 - 2^3 - 3.2^2 - 3.2 - 1$

$2 + 0 + 0 + 4 + 4 - 0 - 0 - 1$

3 ×

$P(3) = 3^7 + 3.3^6 + 3.3^5 + 3^4 - 3^3 - 3.3^2 - 3.3 - 1$

$P(3) = 3 + 3 + 3 + 3 + 3 + 3 + 2$

2

$P(4) = 4^7 + 3.4^6 + 3.4^5 + 4^4 - 4^3 - 3.4^2 - 3.4 - 1$

$P(4) = 4 \quad 0 + 0 + 4 + 2 \quad -1$

3

$P(5) = (-1)^7 + 3.(-1)^6 + 3.(-1)^5 + (-1)^4 - (-1)^3 - 3(-1)^2 - 3(-1) - 1$

$P(5) = 5 + 3 + 3 + 1 + 1 - 3 + 3 - 1$

0

$R_0(P) = \{1, 5\}$

$P = x^7 + 3x^6 + 3x^5 + x^4 - x^3 - 3x^2 - 3x - 1$

$x^4(x^3 + 3x^2 + 3x + 1) - 1(x^3 + 3x^2 + 3x + 1)$

$(x^4 - 1)(x + 1)^3$

$(x^2 + 1)(x^2 - 1)(x + 1)^3$

$(x^2 + 1)(x + 1)(x - 1)(x + 1)^3$

$P = (x + 1)^4 (x^2 + 1)(x - 1) \in \mathbb{Z}_6[x]$

Exercise 31. Consider modular 5 arithmetic from example 16, and the set $S = \{(0,0),(1,1),(2,2),(3,2)\}$. Find a polynomial $P \in \mathbb{Z}_5[x]$ such that $P(x_i) = y_i$ for all $(x_i, y_i) \in S$.

Exercise 32. Consider the set S from the previous example. Why is it not possible to apply algorithm 4 to construct a polynomial $P \in \mathbb{Z}_6[x]$ such that $P(x_i) = y_i$ for all $(x_i, y_i) \in S$?

31 $S = \{(0,0),(1,1),(2,2),(3,2)\}$

32 because 4^{-1} not exist over $\mathbb{Z}_6$.

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2} \cdot \frac{x - x_3}{x_0 - x_3}$$

$$= \frac{x-1}{0-1} \cdot \frac{x-2}{0-2} \cdot \frac{x-3}{0-3}$$

$$= \frac{(x-1)(x-2)(x-3)}{4}$$

$$= (x-1)(x-2)(x-3) \cdot 4$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2} \cdot \frac{x - x_3}{x_1 - x_3}$$

$$= \frac{x}{1-0} \cdot \frac{x-2}{1-2} \quad \frac{x-3}{1-3}$$

$$= \frac{x\,(x-2)(x-3)}{2}$$

$$= x(x-2)(x-3) \cdot 3$$

$$l_2(x) = \frac{x - x_0}{x_2 - x_0} \frac{x - x_1}{x_2 - x_1} \frac{x - x_3}{x_2 - x_3}$$

$$\frac{x-0}{2-0} \cdot \frac{x-1}{2-1} \cdot \frac{x-3}{2-3}$$

$$\frac{x}{2} \cdot \frac{x-1}{1} \cdot \frac{x-3}{-1}$$

$$2 \cdot x(x-1)(x-3)$$

$$l_3(x) = \frac{x - x_0}{x_3 - x_0} \cdot \frac{x - x_1}{x_3 - x_1} \cdot \frac{x - x_2}{x_3 - x_2}$$

$$= \frac{x-0}{3-0} \cdot \frac{x-1}{3-1} \cdot \frac{x-2}{3-2}$$

$$= \frac{x}{3} \cdot \frac{x-1}{2} \cdot \frac{x-2}{1}$$

$$= x(x-1)(x-2)$$

$$P(x) = 0\big((x-1)(x-2)(x-3) \cdot 4\big) + 1 \cdot \big(3x(x-2)(x-3)\big) + 2\big(3x(x-1)(x-3)\big) + 2\big(x(x-1)(x-2)\big)$$

$$= (3x^2 - x)(x-3) + 2\big(2x(x-1)(x-3)\big) + 2x(x-1)(x-2)$$

$$4x(x^2 - 4x + 3) + 2x(x^2 - 3x + 2)$$

$$= (3x^2 - x)(x-3) + 2(2x(x-1)(x-3)) + 2x(x-1)(x-2)$$

$$= 3x^3 + 3x \qquad 4x(x^2+x+3) + 2x(x^2-3x+2)$$

$$= 3x^3 + 3x \qquad 4x^3 + 4x^2 + 12x + 2x^3 + 4x^2 + 4x$$

$$p(x) = 4x^3 + 3x^2 + 4x$$